



Arnav Agrawal

Roll No.:B24CM1012

B.Tech

Artificial Intelligence & Data Science

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Ai-Ds	Indian Institute of Technology, Jodhpur	7.47 (Current)	2024 - Present
12th Grade	CBSE Board	95%	2024

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++, Go, JavaScript, TypeScript, Bash
- **Frameworks & Libraries::** React, Node.js, Express.js, Django, Tailwind CSS, NumPy, Pandas
- **Databases:** MySQL, PostgreSQL, MongoDB, SQLite
- **Tools & Platforms:** Git, Docker, Linux, Postman, Jupyter Notebook, Google Colab

EXPERIENCE

- **Inter IIT Tech Meet 14.0** October 2025 - December 2025
Ebullient Securities Quant High-Prep Problem Statement Team Member GitHub
 - Analyzed **more than 400 anonymized features** on **high-frequency, multi-feature time-series** data to uncover meaningful correlations, patterns, and high-value predictive signals.
 - Developed a custom **vectorized backtester** that evaluated trading signals in **under 20 seconds**, producing risk and return metrics (Sharpe, Calmar, CAGR, drawdown etc.) and interactive plots for trade and strategy analysis.
 - Built **regime-based** trading strategies utilizing engineered indicators, generating **40%+ annualised returns** with **Calmar ratio 14.4**.
- **libr** May 2025 - Present
Founding Engineer GitHub
 - Devised and implemented the platform's **cryptographic communication** layer, establishing **secure key management, content signing, and authenticated communication** across distributed network modules.
 - Migrated content-signing infrastructure from ECDSA to **ed25519**, achieving **2× faster** signing performance and **reducing signature size by 10%**, improving efficiency for distributed content verification.
 - Led **frontend development** and **UI/UX design**, building cross-platform desktop interfaces with **React**.
 - Integrated backend client and moderation services with the React frontend through **Wails**, streamlining cross-layer communication and reducing API integration overhead by nearly **30%**.

PROJECTS

- **Vectorised Backtester** November 2025
A fast backtesting framework for strategy evaluation with realistic constraints and performance analytics. GitHub
- **Orderbook Simulator** November 2025
A simulator for analyzing order matching, execution dynamics, and market microstructure in trading systems. GitHub
 - Implemented **object pooling** and **price-level** primitives to optimize memory reuse, reducing allocation latency by **almost 90%** and increasing simulation throughput to **over 54,000 orders/sec**.
- **Context Clipboard** May 2026
A lightweight, background clipboard manager, that keeps the details of copied stuff for easy revisits. GitHub
 - Built a standalone **C++17** system utility with Windows system APIs for clipboard monitoring, maintaining low footprint (**<20MB RAM** / **<0.1% CPU**) while ensuring **<1ms** history retrieval.
- **DeepState Game** January 2026
A strategy game using graph and tree architectures to model macroeconomic systems and branching narratives. GitHub
 - Co-engineered core logic, reducing complexity from $O(N^2)$ to $O(V + E)$ to accelerate tick processing by **4x**.

COURSEWORK

- Data Structures & Algorithms, Pattern Recognition & Machine Learning, Probability & Statistics, Maths for Computing, Operating Systems, Networking, Digital Design, Computer Architecture, Compiler Design

ACHIEVEMENTS

- **Secured 8th Position** among 23 IITs in the Quant High-Prep PS at Inter IIT Tech Meet 14.0. 2025

CAMPUS INVOLVEMENTS & LEADERSHIP

- **Core Team Member** Quant Club, IIT Jodhpur July 2025 - Present
- **Core Team Member** DevUp Labs - Dev and OSS Community, IIT Jodhpur September 2025 - Present
- **Core Team Member** InSiDe - Game Development Society, IIT Jodhpur August 2025 - June 2026
- **Team Coordinator** Respawn - Gaming and ESports Club, IIT Jodhpur August 2025 - June 2026